

Joyce Yeo Shuhui

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EDUCATION

National University of Singapore

Grad. 2021

B. Comp in Computer Science, Minor in Mathematics (Highest Distinction)

Cum. GPA 4.8/5

- Awarded the full-ride NUS Merit Scholarship and completed the interdisciplinary University Scholars' Programme

Dunman High School

Grad. 2016

Integrated Programme, H3 college-level Mathematics

Full Distinction

EXPERIENCE

Senior Software Engineer (Cloud Infra)

Oct 2022 – Present

TikTok

- Built advanced Kubernetes orchestration capabilities including dual-stack pod networking and interactive webshell terminal features (Golang)
- Fully automated maintenance workflows for hundreds of Kubernetes clusters, by integrating ticketing systems with a custom Argo-based backend (ArgoCD, ArgoWorkflows)
- Slashed business costs by more than 30%, by migrating tens of thousands of microservices to this self-designed cloud environment (BytePlus, AWS, GCP, Aliyun)
- Delivered quarterly presentations and articles on the system's architecture
- Mentored and trained 5 interns and junior engineers

Software Engineer (Engineering Infra)

May 2021 – Sep 2022

Shopee

- Designed the internal tooling that enabled thousands of engineers to easily create data synchronisation pipelines across databases and MQ systems (Golang, MySQL, Kafka, Kubernetes)
- Optimised global database replication throughput by 15x, through compression and batch processing
- Increased product adoption by 20%, by implementing user-defined transformation modules

Applied Cybersecurity Research Assistant

2020 Internship

GovTech Singapore

- Developed a framework to analyse encrypted data using Homomorphic encryption (HE) (C, Python)
- Offered full-time role in a leadership programme

Cybersecurity Research Assistant

2019 Internship

DSO National Laboratories

- Developed counterattacks against anti-debugging features found in malware (C, IDAPro, WinDbg)
- Offered mid-term undergraduate research scholarship

Cryptography Research Assistant

2015 Internship

DSO National Laboratories

- Applied affine transformations to the AES substitution box, and performed linear and differential cryptanalysis against the modified system (C++)
- Awards: 2016 SSEF Merit, offered undergraduate research scholarship

PROJECTS

UltiMafia | MERN (MongoDB, ExpressJS, React, NodeJS)

- Core OSS contributor in this fullstack web game for 2 years
- Closely mentored two high school students based in Seattle and Melbourne

AWARDS

NUS Presidential Honour Roll

2020 NUS Dean's List

2019 DSTA AI Hackathon winner

TECHNICAL SKILLS

Languages: Golang, Python, Javascript, Java

Cloud and DevOps: Kubernetes, Linux, Docker, Ansible, Terraform, Git